

Hanan Gani

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RESEARCH INTERESTS

Intelligent Multi-modal Systems, World/Mental Models, Spatial Reasoning, Embodied AI

EDUCATION

University of California, San Diego (UCSD)

La Jolla, USA

Powell Fellow, Ph.D. in Computer Science · GPA: 4.0/4.0

Sept. 2025 – Present

Advisor: **Manmohan Chandraker** · Fellowships: Powell and Jacobs Engineering Fellowship

Specific focus areas: World models, Mental models, Agentic & Embodied intelligence

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

Master of Science (M.Sc.) in Machine Learning, GPA: 3.77/4.0

Aug. 2022 – May 2024

Advisor: **Salman Khan**, Co-Advisor: **Fahad Khan**

Thesis: Text-to-Image Diffusion with Complex and Detailed Prompts

National Institute of Technology (NIT)

Srinagar, Kashmir, India

Bachelor of Technology (B.Tech), Electronics & Communication Engineering, GPA: 8.561/10

Aug. 2014 – June 2018

Advisor: **G. R. Begh**, Co-Advisor: **Shahid Mehraj**

Thesis: Channel Estimation in Cognitive Radio using Machine Learning

PUBLICATIONS

(* indicates joint first authors, + indicates my role as mentor)

- [NeurIPS'26 under review] **Hanan Gani**, Lulu Shao, and Manmohan Chandraker. “Mental Models for Multi-agent Agents” .
- [ECCV 2026] **Hanan Gani**, Tejal Kulkarni, Madhoolika Chodavarapu, Nicklas Hansen, Manmohan Chandraker. “RoboTALES: Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures”. [Paper] [Code]
- [ACL 2026] Ghazi S. Ahmed, Ahmed Heakl, **Hanan Gani+**, Abdelrahman Shaker, Zhiqiang Shen, Fahad Khan and Salman Khan. “VideoMolmo: Spatio-Temporal Grounding meets Pointing”. [Paper] [Code]
- [ML4RS @ ICLR 2026] **Hanan Gani**, Guy Pulik, Daniel Rosenfeld, Duncan Watson-Parris and Salman Khan. “Recovering Cloud Microstructures with Cascaded Diffusion Inversion.” [Paper] [Code]
- [ICLR 2026] Tajamul Ashraf*, Amal Saqib*, **Hanan Gani**, Muhra AlMahri, Yuhao Li, Noor Ahsan, Umair Nawaz, Jean Lahoud, Hisham Cholakkal, Mubarak Shah, Philip Torr, Fahad Khan, Rao M. Anwer, Salman Khan. “Agent-X: Evaluating Deep Multimodal Reasoning in Vision-Centric Agentic Tasks”. [Paper] [Code]
- [ICCV 2025] Sanjoy Chowdhury*, **Hanan Gani***, Nishit Anand, S. Nag, Ruohan Gao, Mohamed Elhoseiny, Salman Khan, Dinesh Manocha. “AURELIA: Test-time Reasoning Distillation in Audio-Visual LLMs”. [Paper] [Code]
- [CVPR 2025] Shehan Munasinghe*, **Hanan Gani+***, Wenqi Zhu, Jiale Cao, Eric Xing, Fahad Khan, Salman Khan. “VideoGLaMM: A Large Multimodal Model for Pixel-Level Visual Grounding in Videos”. [Paper] [Code]
- [NAACL 2025] **Hanan Gani***, Rohit Bhardwaj*, Muzammal Naseer, Fahad Khan and Salman Khan. “VANE-Bench: Video Anomaly Evaluation Benchmark for Conversational LLMs”. [Paper] [Code]
- [WACV 2025] Raza Imam, **Hanan Gani+**, Mohammad Huzaifa and Karthik Nandakumar. “Test-time Low Rank Adaptation via Confidence Maximization for Zero-Shot Generalization in Vision-Language Models”. [Paper] [Code]
- [COLING 2025] U. Nawaz, Awais M., **Hanan Gani+**, M. Naseer, F. Khan, S. Khan and R. Anwer. “AgriCLIP: Adapting CLIP for Agriculture & Livestock via Domain-Specialized Cross-Model Alignment”. [Paper] [Code]
- [ICLR 2024] **Hanan Gani**, Shariq Farooq, Muzammal Naseer, Salman Khan and Peter Wonka. “LLM Blueprint: Enabling Text-to-Image Generation with Complex and Detailed Prompts”. [Paper] [Code]

- [MICCAI 2024] **Hanan Gani**, Muzammal Naseer, Fahad Khan and Salman Khan. “*MedContext: Learning Contextual Cues for Efficient Volumetric Medical Segmentation*”. [Paper] [Code]
- [NeurIPS 2023] Jameel Hassan*, **Hanan Gani***, N. Hussein, Uzair Khattak, M. Naseer, Fahad Khan, Salman Khan. “*Align Your Prompts: Test-Time Prompting with Distribution Alignment for Zero-Shot Generalization*”. [Paper] [Code]
- [BMVC 2022] **Hanan Gani**, Muzammal Naseer, Mohammad Yaqub. “*How To Train Vision Transformer On Small-scale Datasets?*”. [Paper] [Code]
- [IEEE ICIP 2024] **Hanan Gani**, Nada Saadi, Noor Hussein, Karthik Nandakumar. “*Multi-Attribute Vision Transformers are Efficient and Robust Learners*”. [Paper] [Code]
- [IEEE ICRCV 2018] S. Kumaar, A. Majeedi, **H. Gani**, A. Dogra, Ravi M. Vishwanath, S.N. Omkar. “*A Supervised learning Methodology for Real time Disguised Facial Recognition in Wild*”. [Paper] [Code]

PATENTS

Hanan Gani, Muzammal Naseer, Mohammad Yaqub. “*System and Method of Training Vision Transformer on Small-Scale Datasets*”. [US Patent].

RESEARCH AND INDUSTRY EXPERIENCE

FAIR, Meta – *Research Scientist Intern* Paris, France
Advisor / Manager: Koustav Sinha, World Modeling Team June 2026 – September 2026

University of California, San Diego – *Graduate Student Researcher* California, USA
Advisor: Manmohan Chandraker April 2026 – Present

- Designed a first-order mental-model-enabled framework for social benchmark (under review).
- Exploring spatial intelligence by extending mental models to vision and embodied tasks (ongoing).
- Working on training embodied agents with learnable rewards based on the coherence of future roll-outs (ongoing).

MBZ University of Artificial Intelligence (MBZUAI) – *Research Associate II* Abu Dhabi, UAE
Advisor / Line Manager: Salman Khan June. 2024 – Sept. 2025

- Co-designed **VideoMolmo** for fine-grained point-level grounding; **+5.4 pp** improvement compared to SOTA on proposed VPoS-Bench; open-source code (*Under review*).
- Co-developed **VideoGLaMM** for grounded conversation, visual grounding, & referring segmentation; **+33.74 pp mIoU** improvement over SOTA for GCG task on proposed benchmark; open-source code (*CVPR’25*).
- Developed a super-resolution model for satellite sensory imagery for cloud-seeding ops as a part of collaboration with UAERP department of the UAE Govt. (*ML4RS @ ICLR’26*)
- Developed **VANE-Bench** for LMM inconsistencies in AI videos (*NAACL’25*; released code & eval suite).

King Abdullah University of Science & Tech. (KAUST) – *Visiting Student* Saudi Arabia
Host Advisor: Peter Wonka, Full Professor, CS department June 2023 – Oct. 2023

- Developed **LLM Blueprint** for long, complex T2I diffusion; **69.3% 2-AFC** user preference vs. baselines; open-source release (*ICLR’24*).

MBZ University of Artificial Intelligence (MBZUAI) – *Research Assistant* Abu Dhabi, UAE
Advisors: Mohammad Yaqub, Muzammal Naseer Oct. 2021 – Aug. 2022

- Developed a self-supervised scheme enabling ViTs to train from scratch on small datasets; plug-and-play design; **>90% top-1 acc.** with only 25% train data on CIFAR10 (*BMVC’22, US Patent*).

Fatima Fellowship Predoc Program – *Part-time fellow* USA (remote)
Advisor: Abubakr Abid, Applications Head, Hugging Face Inc March 2021 – Dec. 2021

- Developed a multi-task ViT to efficiently and robustly handle multiple tasks simultaneously within a constrained computational budget; **+59.1%** gain in robust acc. w.r.t baseline under Patch-Fool attack (*IEEE ICIP’24*) .

Samsung-Harman R&D – *Machine Learning Engineer* Bangalore, India (*Oct. 2018 – Sept. 2021*)

- Developed Autoencoder/GAN-based online **anomaly detection** for HMI video streams; **90%** fewer false alarms.
- Developed transformer-based **Recommender Systems** for software fixes (used in production).
- Developed **Log Classifier** to distinguish software & hardware failures; saved **~2 hrs/day** to software team
- Co-developed **Multi-agent UI testing** based on icon detection, screen classifier, OCR (used in production).

Indian Institute of Science (IISc) – *Research Intern* Bangalore, India
Host Lab: Computational Intelligence & UAV Lab, Aerospace Engineering Department Dec. 2017 – March 2018

- Co-developed lightweight disguised-face recognition model; **20 keypoints, 19 FPS** (IEEE ICSIP’18, ICRCV’18).

TECHNICAL SKILLS

Languages: Python, C, C#, MATLAB, SQL

Frameworks: PyTorch, HuggingFace, TensorFlow, Keras, Scikit-Learn, OpenCV

ML Infrastructure & Tooling: Distributed training, DeepSpeed, Accelerate, PyTorch Lightning, W&B, TensorBoard

Dev Tools: Linux, Git, Google Cloud Platform, VS Code, PyCharm, Linux, Flask

TEACHING AND ACADEMIC SERVICES

Teaching | MBZUAI

- TA for **CV806** (Advanced Topics in Vision & Language, Spring'24), **AI702** (Deep Learning, Spring'24), **ML703** (Probabilistic & Statistical Inference, Fall'23), **ML701** (Machine Learning, Fall'23).
- Tutor & Lab Instructor for **UGRIP program** at MBZUAI (Summer'24).

Conference Services

- **Regular Reviewer:** NeurIPS, ICLR, CVPR, ICCV, ECCV, ICML, AAAI
- **Program Volunteer:** IEEE ICIP'24 (Abu Dhabi, UAE)

Invited Talks

- **Abu Dhabi University**, May'25: Delivered a talk on "Latest Trends in AI" at CS department.
- **University of Jordan**, June'24: Presented a talk on "Generative AI" to undergrad CS students.

Mentorship

- Mentoring 2 MS student at UCSD on embodied agents, collaboration resulted in a paper at ECCV'26.
- Mentoring an undergraduate student at UCSD on the topic of spatial intelligence.
- Mentored 3 MS students in thesis and publication guidance via MBZUAI Pilot Mentorship Program (2024-2025).

HONORS AND AWARDS

- Recipient of prestigious **Powell Fellowship** and **Jacobs Engineering Fellowship** at UC San Diego.
- 2nd place in Research track at UC Berkeley's **AgentX Competition**.
- Awarded **ICCV 2025** Travel grant.
- Awarded **CVPR 2025** Travel Award
- Honorable mention by MBZUAI for my research contributions during masters degree ([link](#))
- Awarded **ICLR 2024** Travel Grant
- Awarded **NeurIPS 2023** Travel Grant
- MBZUAI graduate studies scholarship holder
- Received Samsung-Harman Star Excellence award from India Regional Head
- Merit Based Scholarship granted for undergraduate studies by Ministry of Minority Affairs India.

EXTRACURRICULAR AND SOCIAL ACTIVITIES

Graduate Student Council | Machine Learning Coordinator; liaison with department leadership; student initiatives.

Rivero | Co-Founder; NGO guiding students in Kashmir; workshops and counseling for ~2,000+ underprivileged students